

Not All Scores Are Probabilities: Typed, Calibrated Confidence Contracts for Composed Agent Systems

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Abstract—Composed agent systems pass around many numbers in $[0, 1]$: a model’s self-reported confidence, an embedding similarity, a weighted risk indicator, a posterior mean, a selection propensity. They look interchangeable, and they are routinely thresholded, averaged and multiplied as if they were probabilities. We propose a typed contract. Every emitted score declares its kind (*probability, posterior mean, score, similarity or risk indicator*), its source, its range, whether and against what it is calibrated, and the code that produces it. Five rules then govern combination. We prove the premises of four of the five rules (R4 is a design requirement): averaging or taking the maximum across producers is not invariant under monotone reparametrization, so the combined ranking can flip while every input’s ranking is unchanged; products of calibrated, conditionally independent probabilities are calibrated for the conjunction; isotonic regression makes a score calibrated in-sample; and substituting defaults for absent values biases every mean toward the default. Across three systems we find 13 declared scores. Exactly one is a calibrated probability, and none of the 78 pairs of scores (50 of them across systems) may be combined numerically. The contract is enforced by a conformance test in each system’s repository. On a synthetic overconfident score, an isotonic fit reduces held-out expected calibration error (ECE) from 0.277 to 0.018.

Index Terms—calibration, confidence scores, measurement scales, isotonic regression, composed AI systems, software contracts

I. INTRODUCTION

Consider three of our systems. ContextWeave’s retrieval router was rewarded with a model’s self-confidence [1] until a verified reward replaced it [4]; the self-confidence is still emitted and logged. DeviceWeave executes a command when $\alpha \cdot \text{cosine} + (1 - \alpha) \cdot \text{behavior} \geq 0.4$ [2]. CipherWeave overrides to post-quantum encryption when a maximum of standardized deviations exceeds 3 [3], [5]. Each number lives in $[0, 1]$, or looks like it does, and each means something different. When systems compose, these numbers meet. Nothing in their type says whether 0.7×0.8 or $\frac{1}{2}(0.7 + 0.8)$ means anything.

II. THE CONTRACT

Definition 1 (Score envelope). *A score is emitted as $\langle \text{value}, \text{observed}, \text{name}, \text{kind}, \text{source}, \text{calibrated}, \text{event?}, \text{evidence?}, \text{neutral?} \rangle$. kind is one of:*

- *probability*: $P(\text{event})$, and must be calibrated against that event;
- *posterior_mean*: of a stated quantity, carrying its evidence count;
- *score*: ordering-only;

- *similarity*: its scale belongs to the model that produced it, and it declares its neutral point;
- *risk_indicator*: a standardized or weighted indicator, possibly unbounded, carrying its evidence count.

An unobserved score has value null and observed false.

Rules.

- R1 Only probabilities compose by probability rules.
- R2 A probability must be calibrated, and must name the event it is calibrated against.
- R3 Never average, add or max scores of different kinds or producers; combine *decisions* instead. Within one producer, whose outputs share a scale by construction, these operations are allowed; for example, a drift detector may take the maximum of its own standardized channels.
- R4 Carry the evidence count when reliability depends on it.
- R5 Absence is not zero.

Each repository declares its scores in a registry that names the producing function. A conformance test (`tests/test_score_contract.py` in each repository) checks the registry against the contract, imports every producer (so a declaration cannot outlive its code), and pins the contract file’s hash, so vendored copies cannot drift. The checker `can_combine(a, b, op)` names the rule that forbids an operation.

III. WHY THE RULES HOLD

Proposition 1 (Cross-producer averages are not ordinal). *Let u, v be scores from different producers, each meaningful only up to a strictly increasing transformation. For any $\alpha \in (0, 1)$ there exist candidates A, B and a strictly increasing f with $\alpha u_A + (1 - \alpha)v_A < \alpha u_B + (1 - \alpha)v_B$ but $\alpha f(u_A) + (1 - \alpha)f(u_A) > \alpha f(u_B) + (1 - \alpha)f(u_B)$.*

Proof. Take $u_A > u_B$ and $v_A < v_B$ with $\alpha(u_A - u_B) < (1 - \alpha)(v_B - v_A)$, so that B ranks higher. Choose f strictly increasing with $f(u_A) - f(u_B) > \frac{1 - \alpha}{\alpha}(v_B - v_A)$, for example a steep enough power. Then A ranks higher. Neither input’s own ordering has changed. $\square$

Concretely, write each candidate as (similarity u , behavior score v) and take $\alpha = \frac{1}{2}$, $A = (0.90, 0.10)$ and $B = (0.60, 0.50)$: the raw averages rank B first (0.550 vs. 0.500). Cubing the similarity u , a monotone change that could come from simply switching embedding models, ranks A first (0.414 vs. 0.358). This is DeviceWeave’s `final_score` pattern.

The same construction applies to $\max$, by choosing f to lift u_A above v_B .

Proposition 2 (Calibrated products are calibrated). *If q_1, q_2 are calibrated probabilities for events E_1, E_2 , i.e. $P(E_i | q_i) = q_i$, and E_1, E_2 are conditionally independent given (q_1, q_2) , with $P(E_i | q_1, q_2) = P(E_i | q_i)$, then $P(E_1 \wedge E_2 | q_1, q_2) = q_1 q_2$.*

Proof. By conditional independence, $P(E_1 \wedge E_2 | q_1, q_2) = P(E_1 | q_1, q_2) P(E_2 | q_1, q_2) = q_1 q_2$. $\square$

Both hypotheses are needed, which is why R1 requires the kind to be *probability* and R2 requires calibration. Independence remains the caller’s claim.

Proposition 3 (Isotonic calibration). *Let $\hat{g}$ be the pool-adjacent-violators (PAV) fit to pairs (s_j, o_j) sorted by s . Then $\hat{g}$ is the least-squares non-decreasing fit [6]. On each constant block $\mathcal{B}$ of $\hat{g}$, the fitted value equals the block’s mean outcome, so the calibration error of $\hat{g}$ over its own blocks is zero in-sample.*

Proof. PAV merges adjacent blocks while their means violate monotonicity, and replaces each merged block by its mean. It is standard that this produces the L^2 projection onto the cone of monotone sequences [6]. Each block’s fitted value is by construction the mean of its outcomes, which is the in-sample calibration statement. $\square$

Example. We drew 5,000 synthetic outcomes, 60% correct. The score was $\mathcal{N}(0.9, 0.05)$ for a correct outcome and $\mathcal{N}(0.8, 0.05)$ for an incorrect one, clipped to $[0, 1]$. The two distributions overlap, and both sit far above the 60% base rate, so the score is overconfident. We fit the isotonic map on the first 2,500 samples and evaluated it on the other 2,500, measuring ECE with 10 equal-width bins. ECE fell from 0.277 to 0.018, and Brier score from 0.258 to 0.108.

Proposition 4 (Defaults bias means). *If a score with mean μ is missing with probability m (independently of its value), and a default κ is substituted, the observed mean is $(1-m)\mu + m\kappa$. Two sources with equal μ but different m then appear different. Dropping missing values recovers μ .*

Proof. This follows from total expectation over the missingness indicator, and from the independence of missingness and value. $\square$

IV. CENSUS

We declared every score emitted by three systems (Table I).

Sources: 6 measurement, 3 heuristic, 2 self-reported, 1 model judge and 1 human. The only calibrated probability is the router’s selection propensity. The quantity it estimates, the chance that Thompson sampling picks the arm given the current posteriors, is a genuine probability by definition. The system estimates it by Monte Carlo with 200 draws, a standard error of at most 0.035. Of the 78 pairs of distinct scores, 50 of which cross system boundaries, `can_combine` allows *none* to be averaged, multiplied or maxed. Wherever these systems

TABLE I
DECLARED SCORES BY KIND AND SYSTEM.

System	prob.	post.	score	sim.	risk
ContextWeave	1	1	5	0	0
DeviceWeave	0	0	3	1	0
CipherWeave	0	0	0	0	2
Total (13)	1	1	8	1	2

compose, they must combine decisions (“both gates pass”), not numbers.

V. RELATED WORK

Calibration of classifiers and neural networks is surveyed by Guo et al. [7]. Isotonic and histogram calibration follow Zadrozny and Elkan [8]. Measurement theory’s distinction between ordinal and interval scales [9] is the formal root of R3.

VI. CONCLUSION

A number in $[0, 1]$ is not a probability until someone has checked it against outcomes. Typing each score, and refusing combinations its type does not license, turns a silent category error into a failing test.

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